

CHEM 231 — Exam 1 Practice (Klein Ch 1–4)

Maps to your first real course exam, ~week of Sep 15. Build this into your semester rhythm, not this weekend.

Scope — from Prof. Ciesla's schedule, Exam 1 falls after Chapters 1–4: **Ch 1** (gen-chem review — the prereq material), **Ch 2** (molecular representation: bond-line, functional groups), **Ch 3** (acids & bases), **Ch 4** (alkanes & cycloalkanes: nomenclature, conformations). This practice exam covers all four. **Take it in September, after those chapters are taught** — right now Ch 2–4 haven't been covered, so treat sections C–D as a preview of what's coming, not a this-weekend task.

Format: blank-page, ~90 min, no notes. Draw everything. Photograph and send to Claude for a scored gap report mapped to the specific Klein sections and the day's lecture.

Section A • Chapter 1 review (structure, charge, geometry) you know this now

A1. Draw the Lewis structure of the nitrate ion, NO_3^- , showing all formal charges. How many equivalent resonance structures does it have?

6 pts

A2. For each atom in HCN, give its formal charge and hybridization. Is the molecule polar?

6 pts

A3. Rank by boiling point, highest first, with a one-line reason: propane, ethanol, dimethyl ether.

5 pts

Section B • Chapter 2 — molecular representation taught week 2

B1. Convert this condensed formula to a bond-line (skeletal) structure: $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CH}_2\text{OH}$.

5 pts

B2. A bond-line structure has 6 vertices/ends in a chain with a $=\text{O}$ on the second carbon. How many hydrogens are on carbon 1? On carbon 2? State the rule you used.

5 pts

B3. Identify the functional group in each: (a) R-OH , (b) R-O-R , (c) R-NH_2 , (d) R-CHO , (e) R-CO-R , (f) R-COOH , (g) R-COO-R .

7 pts

B4. Compute the degrees of unsaturation for $\text{C}_5\text{H}_8\text{O}$. What structural features could account for them?

5 pts

Section C · Chapter 3 — acids & bases

taught week 3

C1. Define a Brønsted–Lowry acid and base. In $\text{HCl} + \text{H}_2\text{O} \rightarrow \text{Cl}^- + \text{H}_3\text{O}^+$, label each species' role and identify the conjugate acid–base pairs.

6 pts

C2. Lower pK_a means a stronger or weaker acid? If acid HA has pK_a 5 and HB has pK_a 16, which is the stronger acid, and which has the more stable conjugate base?

5 pts

C3. Using ARIIO (Atom, Resonance, Induction, Orbital), explain which is more acidic and why: (a) ethanol vs acetic acid, (b) HF vs HCl, (c) CH_3COOH vs CF_3COOH .

9 pts

C4. In an acid–base reaction, does equilibrium favor the side with the weaker or stronger acid? Predict the direction: acetate (CH_3COO^-) + water.

5 pts

D1. Give the IUPAC name: a 5-carbon chain with a methyl group on C2 and a methyl group on C3. ^{5 pts}

D2. Draw all constitutional isomers of C_5H_{12} and name each.

6 pts

D3. Draw a Newman projection of butane (sighting $C2 \rightarrow C3$) in its anti and its fully eclipsed conformations. Which is lower energy and why?

6 pts

D4. For methylcyclohexane in a chair: does the methyl prefer axial or equatorial? Explain using 1,3-diaxial interactions.

5 pts

D5. Rank ring strain, highest to lowest: cyclopropane, cyclohexane, cyclobutane. One-line why cyclohexane is the odd one out.

4 pts

■ STOP ■

Answer key on the next page. Take the whole exam cold first.

Answer Key

Section A — Ch 1 review

A1. NO_3^- : N center, double-bonded to one O and single-bonded to two O^- . N carries FC +1; the two single-bonded O's each -1; double-bonded O is 0. Net -1. **3 equivalent resonance structures** (the double bond rotates among the three O's); real bond order 1.33.

A2. $\text{H}-\text{C}\equiv\text{N}$: H (FC 0), C (FC 0, **sp**), N (FC 0, **sp**, one lone pair). Molecule is **polar** (linear but asymmetric; N end is δ^-).

A3. ethanol (H-bonds, O-H) > dimethyl ether (dipole-dipole, no O-H) > propane (dispersion only, nonpolar).

Section B — Ch 2

B1. A 4-carbon zig-zag chain: $\text{C1}-\text{C2}-\text{C3}-\text{C4}$, with a CH_3 branch on C3 and an -OH on C4 (drawn as a line ending in OH). (2-methylbutan-1-ol skeleton.)

B2. Rule: each carbon fills to 4 bonds with implied H's. C1 (chain end, 1 bond shown) = **3 H**. C2 with a $\text{C}=\text{O}$ (2 bonds to O + 1 to C1 + 1 to C3 = 4 already) = **0 H** (it's a carbonyl carbon).

B3. (a) alcohol, (b) ether, (c) amine, (d) aldehyde, (e) ketone, (f) carboxylic acid, (g) ester.

B4. $\text{DoU} = (2 \cdot 5 + 2 - 8)/2 = 2$. Could be two double bonds, one triple bond, one ring + one double bond, or two rings (O doesn't change the formula count).

Section C — Ch 3

C1. Acid = proton donor, base = proton acceptor. HCl = acid, H_2O = base; conjugate pairs: HCl/Cl^- and $\text{H}_2\text{O}/\text{H}_3\text{O}^+$.

C2. Lower pK_a = **stronger** acid. HA (pK_a 5) is stronger; the stronger acid has the **more stable conjugate base** (A^-).

C3. (a) Acetic acid — its conjugate base (acetate) is **resonance**-stabilized over two O's; ethoxide is not (R = Resonance). (b) HCl — bigger atom (Cl) holds the negative charge better than F (A = Atom, size down a column). (c) CF_3COOH — the electron-withdrawing F's stabilize the conjugate base by **induction** (I = Induction).

C4. Equilibrium favors the side with the **weaker** acid (more stable). Acetate + water: acetate (conj. base of pK_a -5 acid) is a weak base; water (pK_a 15.7) is a far weaker acid than acetic acid, so equilibrium lies **far left** — acetate does not appreciably deprotonate water.

Section D — Ch 4

D1. 2,3-dimethylpentane.

D2. Three isomers: **pentane** (straight), **2-methylbutane** (isopentane), **2,2-dimethylpropane** (neopentane).

D3. Anti = the two methyls 180° apart (staggered) — **lowest energy** (minimal steric strain, no torsional strain). Fully eclipsed = methyls overlapping at 0° — highest energy (steric + torsional strain).

D4. Equatorial. An axial methyl suffers 1,3-diaxial steric clashes with the two axial H's on the same face; equatorial points away from the ring, avoiding them. ~95% sits equatorial.

D5. cyclopropane > cyclobutane > cyclohexane. Cyclohexane is strain-free because it adopts a puckered **chair** with ideal 109.5° angles and all-staggered bonds; the small rings can't.

Exams 2 & 3 — build them when you get there.

Exam 2 (~Oct 20) covers Ch 5 (stereoisomerism: chirality, R/S, enantiomers/diastereomers), Ch 6 (chemical reactivity & mechanisms: arrow-pushing, nucleophile/electrophile), Ch 7 (substitution: S_N1/S_N2).

Exam 3 (~Nov 10) covers Ch 8 (alkenes: structure, elimination E1/E2) and Ch 9 (addition reactions of alkenes).

These are reaction-and-mechanism heavy — I can write accurate practice exams for each once you've been taught the chapter. When you reach it, bring the chapter (or just tell me the sections) and I'll generate that exam the same way.

Building them now would mean drilling mechanisms you haven't seen — not useful yet.